

Limit-Request Combination Table (Kubernetes)

This table represents the exact Kubernetes admission-time behavior for requests and limits, with and without ResourceQuota, including resulting QoS classes.

Case 1: ResourceQuota EXISTS

Requests Set	Limits Set	Request $\leq$ Limit?	Within Namespace Quota?	Result	What Happens
Yes	Yes	Yes	Yes	Pod runs	Normal scheduling
Yes	Yes	No	-	Rejected	Request must be $\leq$ limit
Yes	No	-	Yes	Rejected	Quota requires limits
No	Yes	-	-	Rejected	Quota requires requests
No	No	-	-	Rejected	Pod creation blocked
Yes	Yes	Yes	No	Rejected	Exceeded namespace quota

Rule: If ResourceQuota is defined, required resource fields (requests and/or limits) must be explicitly present or injected via LimitRange.

Case 2: ResourceQuota does NOT exist

Requests Set	Limits Set	Result	What Happens
No	No	Pod runs	BestEffort Pod
No	Yes	Pod runs	Burstable Pod
Yes	No	Pod runs	Burstable Pod
Yes	Yes	Pod runs	Guaranteed Pod

Kubernetes QoS Classes (Interview Favorite)

Requests	Limits	QoS Class
No	No	BestEffort
Yes	No	Burstable
No	Yes	Burstable
Same	Same	Guaranteed

Kubernetes LimitRange & ResourceQuota – All-in-One Printable Cheat Sheet

This document summarizes how LimitRange and ResourceQuota behave together in Kubernetes, including all combinations, QoS mapping, and admission evaluation order. Designed for quick revision, interviews, and production reference.

Evaluation Order (Admission Time)

Step	Component	What Happens
1	LimitRange	Validation (min/max rules)
2	LimitRange	Default injection (requests/limits)
3	ResourceQuota	Namespace total validation
4	API Server	Pod admitted or rejected

Case 1: ResourceQuota EXISTS + LimitRange EXISTS

Requests Set	Limits Set	Defaults Injected?	Within Quota?	Result	Explanation
No	No	Yes	Yes	Pod runs	LimitRange injects defaults, quota validates
No	No	No	-	Rejected	Quota requires requests
Yes	No	Maybe	Yes	Allowed / Rejected	Depends on default limits and quota rules
No	Yes	Maybe	No	Rejected	Quota requires requests
Yes	Yes	-	Yes	Pod runs	Fully valid
Yes	Yes	-	No	Rejected	Namespace quota exceeded
Yes	Yes	-	-	Rejected	Request > Limit (LimitRange failure)

Case 2: ResourceQuota EXISTS + LimitRange DOES NOT exist

Requests Set	Limits Set	Result	Explanation
No	No	Rejected	Quota cannot calculate usage
No	Yes	Rejected	Quota requires requests
Yes	No	Allowed / Rejected	Allowed if quota does not track limits
Yes	Yes	Allowed / Rejected	Allowed only if within quota

Case 3: LimitRange EXISTS + ResourceQuota DOES NOT exist

Requests Set	Limits Set	Result	QoS	Explanation
No	No	Pod runs	Burstable	Defaults injected
No	No	Rejected	-	Minimum values violated
No	Yes	Pod runs	Burstable	Max enforced
Yes	No	Pod runs	Burstable	Valid config

Requests Set	Limits Set	Result	QoS	Explanation
Yes	Yes	Pod runs	Guaranteed	Ideal config

Case 4: NEITHER ResourceQuota NOR LimitRange exists

Requests Set	Limits Set	Result	QoS	Risk
No	No	Pod runs	BestEffort	Very High
No	Yes	Pod runs	Burstable	Medium
Yes	No	Pod runs	Burstable	Medium
Yes	Yes	Pod runs	Guaranteed	Low

QoS Class Mapping

Requests	Limits	QoS Class
No	No	BestEffort
Yes	No	Burstable
No	Yes	Burstable
Same	Same	Guaranteed

Golden Rules (Remember These)

- LimitRange controls per-container correctness and defaults
- ResourceQuota controls namespace-wide totals
- LimitRange runs BEFORE ResourceQuota
- ResourceQuota checks requested values, not actual usage
- Use both together in production

Kubernetes LimitRange & ResourceQuota – Cheat Sheet

This cheat sheet explains how LimitRange and ResourceQuota behave together in Kubernetes, including evaluation order and common interview traps.

Evaluation Order (Admission Time)

1. LimitRange validation
2. LimitRange default injection
3. ResourceQuota validation
4. Pod admitted

Case 1: ResourceQuota EXISTS + LimitRange EXISTS

Requests	Limits	Defaults Injected	Within Quota	Result
No	No	Yes	Yes	Pod Runs
No	No	No	-	Rejected
Yes	No	Maybe	Yes	Allowed / Rejected
No	Yes	Maybe	No	Rejected
Yes	Yes	-	Yes	Pod Runs
Yes	Yes	-	No	Rejected

Case 2: ResourceQuota EXISTS + LimitRange NOT EXISTS

Requests	Limits	Result
No	No	Rejected
No	Yes	Rejected
Yes	No	Allowed / Rejected
Yes	Yes	Allowed / Rejected

Case 3: LimitRange EXISTS + ResourceQuota NOT EXISTS

Requests	Limits	Result	QoS
No	No	Pod Runs	Burstable
No	Yes	Pod Runs	Burstable
Yes	No	Pod Runs	Burstable
Yes	Yes	Pod Runs	Guaranteed

Case 4: Neither LimitRange nor ResourceQuota Exists

Requests	Limits	Result	QoS	Risk
No	No	Pod Runs	BestEffort	High
No	Yes	Pod Runs	Burstable	Medium
Yes	No	Pod Runs	Burstable	Medium
Yes	Yes	Pod Runs	Guaranteed	Low

QoS Class Mapping

Requests	Limits	QoS
No	No	BestEffort
Yes	No	Burstable
No	Yes	Burstable
Same	Same	Guaranteed